International Conference on Power, Electronics, Communications, Computing, and Intelligent Infrastructure 2026 (PECCII), 17-18 June 2026, PUST, Pabna, Bangladesh

2026 International Conference on Power, Electronics, Communications, Computing, and Intelligent Infrastructure (PECCII) | 979-8-3195-3395-1/26/$31.00 ©2026 IEEE | DOI: 10.1109/PECCII70991.2026.11662065

Secure Outage Performance Analysis of Reconfigurable Intelligent Surface-Aided Networks with Selection Combining Diversity Scheme

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***Abstract*—This paper investigates the outage-based secrecy performance of reconfigurable intelligent surface (RIS)-assisted wireless communication systems with selection combining (SC) diversity over Rayleigh fading channels in the presence of a passive multi-antenna eavesdropper. Unlike existing RIS-assisted secrecy studies that mainly focus on single-antenna reception without diversity combining, this work considers a single-antenna source communicating with a multi-antenna destination through an RIS, while a passive multi-antenna eavesdropper attempts to intercept the transmission. Since the direct links between the source and both the destination and the eavesdropper are blocked by physical obstacles, all communications are established through an RIS equipped with N passive reflecting elements. Exact closed-form analytical expressions for the probability of non-zero secrecy capacity (PNZSC), intercept probability (IP), and secure outage probability (SOP) are derived and verified through Monte Carlo simulations. Numerical results demonstrate that increasing the number of RIS reflecting elements N and legitimate receiver antennas N_d significantly improves secrecy performance, whereas increasing the number of eavesdropper antennas N_e degrades secrecy reliability. Furthermore, RIS deployment can partially compensate for stronger eavesdropping conditions without increasing transmit-side complexity. The obtained results confirm that integrating RIS technology with low-complexity SC diversity provides an effective and energy-efficient solution for enhancing physical-layer security in future wireless networks.**

***Index Terms*—Reconfigurable intelligent surface (RIS), selection combining (SC), physical layer security (PLS), intercept probability (IP), probability of non-zero secrecy capacity (PNZSC), secure outage probability (SOP).**

I. Introduction

Wireless communication systems are naturally susceptible to interception due to the open and broadcast characteristics of the wireless medium. Physical layer security (PLS) addresses the eavesdropping threat by exploiting the inherent randomness of wireless channels to achieve information-theoretic secrecy without relying on secret key distribution. Recently, RIS has been proposed as a promising alternative to ensure reliability, security, and energy optimization by intelligently reflecting signals [1]. Cognitive Radio (CR) networks further enhance spectrum utilization by allowing secondary users to share authorized spectrum under interference constraints. The integration of RIS with CR networks is investigated in [2], which proposes a RIS-assisted CR framework to improve the throughput of the secondary network while enhancing the secrecy performance of the primary network. The study further presents analytical derivations of the secrecy outage probability under practical diversity combining schemes.

A. Related Works

Recently, Ashutosh et al. [3] analyzed the secrecy behavior of RIS-supported wireless networks and derived closed-form expressions for SOP, IP, and PNZSC. Their findings showed that secrecy diversity improves with an increasing number of RIS reflecting elements. However, their analysis was limited to a single legitimate receiver and did not take advantage of receiver-side diversity combining methods, restricting secrecy reliability in fading channels. Similarly, L. Yang et al. [4] derived analytical expressions for SOP and PNZSC in RIS-assisted systems, but their study was limited to a single-antenna reception model and did not account for diversity combining. Yang et al. [5] investigated the average capacity of RIS-assisted wireless-powered relay networks with energy constraints and proposed mode selection to improve average performance. Their work provided valuable insights into RIS-assisted relaying, but focused on ergodic capacity rather than physical layer security. Cui et al. [6] assessed the coverage probability based on SNR in RIS-assisted wireless communications, which provides insights into reliability improvements with RIS deployment; however, physical layer security issues were not considered. Do et al. [7] explored the secure performance of RIS-assisted networks with hardware impairments, focusing on SOP and average secrecy rate, but did not include PNZSC, intercept probability, or diversity combining. The impact of eavesdropper mobility was considered in [8], where the ergodic secrecy capacity under RIS-based transmission was analyzed in the presence of randomly moving UAV-based eavesdroppers. Despite realistic modeling of eavesdropping scenarios, this work was limited to ergodic metrics and did not evaluate SOP, PNZSC, or intercept probability. Basharat et al. [9] explored RIS applications and challenges, including integration with emerging 6G network technologies, but did not provide analytical evaluations of secrecy. Similarly, Basar et al. established the theoretical framework for reconfigurable intelligent surfaces (RIS) by demonstrating the well-known N² signal-to-noise ratio (SNR) improvement and controllable propagation conditions [10]. Nevertheless, these foundational

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TABLE I
COMPARISON WITH EXISTING WORKS

Ref.	Channel	RIS	SC Diversity	Security	Multi-Antenna	Metrics
[3]	Rayleigh	✓	×	✓	×	SOP, IP, PNZSC
[4]	Rayleigh	✓	×	✓	×	SOP, PNZSC
[5]	Rayleigh	✓	×	×	×	EC
[6]	Rayleigh	✓	×	×	×	CP
[7]	Rayleigh	✓	×	✓	×	SOP, ASR
[8]	Rayleigh	✓	×	✓	×	ESC
[2]	Rayleigh	✓	×	✓	×	SOP, OP
[11]	Rayleigh	×	✓	✓	✓	SOP, ASR
This work	**Rayleigh**	✓	✓	✓	✓	**IP, PNZSC, SOP**

studies did not focus on secrecy performance metrics or the impact of receiver diversity combining. Although relay-aided and jamming-based systems employing SC and MRC were analyzed in [11], these systems rely on active, energy-intensive relays and do not take advantage of passive propagation control enabled by RIS. Even with the increasing literature, a comprehensive analysis of the outage-based secrecy performance of RIS-enhanced wireless networks employing diversity-based receiver-side selection has been lacking—an analysis that is crucial for understanding and improving security measures in these systems. In particular, the combined evaluation of SOP, PNZSC, and intercept probability under Rayleigh fading conditions, while considering SC diversity at the receiver and eavesdropper, has been scarcely explored. Recognizing these gaps, this study aims to characterize the secrecy reliability of RIS-aided systems under realistic fading conditions, employing low-complexity SC diversity.

B. Contributions

Considering the limitations of existing studies and the increasing demand for reliable and secure wireless communications, this work develops an analytical model to assess outage-related secrecy performance in RIS-enabled wireless networks employing selection combining (SC) diversity under Rayleigh fading conditions.

- **I**nitially, closed-form equations for Intercept Probability (IP), PNZSC, and SOP are obtained for RIS-assisted wireless networks employing SC diversity operating over Rayleigh fading channels.
- **T**he effect of key system parameters, including the number of RIS reflecting elements and the average SNR at the eavesdropper, on the considered secrecy metrics (IP, PNZSC, and SOP) is examined in detail.
- **T**he effect of the antenna configuration at both the legitimate receiver and the eavesdropper on PNZSC and SOP is investigated.
- **L**astly, analytical equations are verified by Monte Carlo simulations, which demonstrate close agreement between the analytical and simulation results.

The remaining part of the paper is arranged as follows. The system framework and channel assumptions are introduced in Section II. Section III develops the analytical expressions for the secrecy metrics. Section IV provides numerical evaluations along with simulation validation, and Section V concludes the study.

II. SYSTEM MODEL

A. Network Configuration and Channel Environment

A secure wireless communication framework is considered in which the source (S) is equipped with a single antenna, the destination (D) has N_d antennas, and the eavesdropper (E) is equipped with N_e antennas. In this system, the source–receiver links (S–D and S–E) are blocked by a physical obstacle such as a building, for which a RIS can be employed to facilitate signal reflection as shown in Fig. 1.

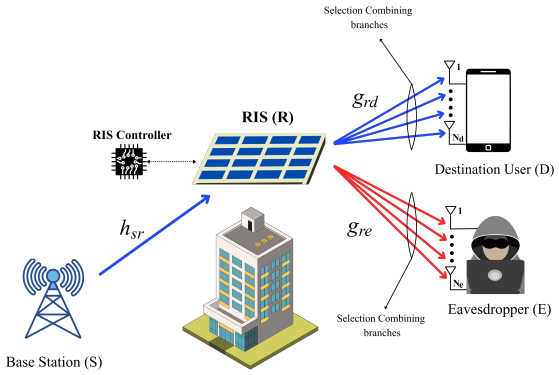


Fig. 1. System model of the proposed scheme.

The reconfigurable intelligent surface (RIS) R includes N reflecting elements and a microcontroller. Each reflecting element can change the phase of the received signal for constructive interference at the legitimate receiver. We consider a block fading channel model for which all channels are Rayleigh distributed without a Line-of-Sight (LoS) component due to the urban landscape.

B. Signal and Channel Modeling

The communication process includes two hops in order to overcome physical barriers. In the first hop, the source S sends a signal x with $E[|x|^2] = 1$, which ensures that the signal has a consistent power level. The channel coefficients between the source S and RIS R with N elements are represented by $\mathbf{h}_{sr} \in \mathbb{C}^{N\times 1}$, where $\mathbf{h}_{sr} = [h_1, h_2, \ldots, h_N]^T$.

In the second hop, the RIS reflects the incident signal towards the receiver. The RIS phase-shift matrix is represented as:

$$\mathbf{\Theta} = \text{diag}([e^{j\phi_1}, e^{j\phi_2}, \ldots, e^{j\phi_N}]) \tag{1}$$

The phase-shift matrix Θ in (1) is a diagonal matrix where each diagonal entry $(e^{j\phi_i})$ represents a unit-magnitude complex reflection coefficient applied by the i-th RIS element. The phase $(\phi_i \in [0, 2\pi])$ is optimized by the RIS controller to align the reflected signals constructively at the destination, thereby maximizing the received SNR at D.

$$y_{D,k} = \sqrt{P_s}(\mathbf{g}_{rd,k}^H \mathbf{\Theta} \mathbf{h}_{sr})x + z_{D,k} \tag{2}$$

$$y_{E,t} = \sqrt{P_s}(\mathbf{g}_{re,t}^H \mathbf{\Theta} \mathbf{h}_{sr})x + z_{E,t} \tag{3}$$

In (2) and (3), P_s denotes the transmission power. The term $(g_{rd,k}^H \Theta h_{sr})$ represents the effective cascaded channel gain

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from the source S to the k-th antenna of the destination D (or the t-th antenna of the eavesdropper E) through the RIS. This cascaded channel incorporates both the source-to-RIS channel (h_{sr}) and the RIS-to-receiver channel $(g_{rd,k})$, while the RIS phase-shift matrix Θ adaptively adjusts the reflected signals to achieve constructive combining at the receiver. The noise terms $(z_{D,k})$ and $(z_{E,t})$ are modeled as additive white Gaussian noise (AWGN) with zero mean and variance σ^2.

C. Selection Combining (SC) Diversity

A low-complexity SC scheme is adopted at the legitimate receiver as well as the eavesdropper to support reliable secure communication. In this technique, the legitimate receiver monitors all available diversity branches and selects the branch with the maximum instantaneous SNR. Let the instantaneous SNR at the k-th branch of the destination be $\gamma_{D,k}$. The output SNR after SC is expressed as:

$$\gamma_D^{SC} = \max_{k\in\{1,\ldots,N_D\}}(\gamma_{D,k}) \tag{4}$$

III. Problem Formulation

A. PDF of the RIS-aided networks for destination user and eavesdropper

The probability density functions (PDFs) of the received SNRs at the legitimate receiver and the eavesdropper are given in (5) and (6), respectively. In these expressions, γ_d and γ_e denote the instantaneous SNRs of the main and eavesdropper channels, respectively. Furthermore, ω_d and ω_e denote the average SNRs of the corresponding channels.

The interference-plus-noise terms of the respective channels are assumed to follow exponential distributions with parameters Λ_d and Λ_e. In the summation expressions, the indices k and t vary over the SC diversity branches. Moreover, the binomial coefficients $\binom{N_d-1}{k}$ and $\binom{N_e-1}{t}$ appear in the respective summations. The PDFs of the received SNRs at the destination user are given by [11] eq. (31)

$$f_{\gamma_d}^{\mathrm{SC}}(\gamma_d) = \sum_{k=0}^{N_d-1} \frac{N_d(-1)^k}{\omega_d\Lambda_d}\binom{N_d-1}{k}\frac{1+\frac{1}{\Lambda_d}+\frac{(k+1)\gamma_d}{\omega_d}}{\left(\frac{1}{\Lambda_d}+\frac{(k+1)\gamma_d}{\omega_d}\right)^2} \times \exp\left(-\frac{(k+1)\gamma_d}{\omega_d}\right) \tag{5}$$

where the constants are defined as

$$A_k = \frac{N_d(-1)^k}{\omega_d\Lambda_d}\binom{N_d-1}{k}, \qquad B_d = \frac{1}{\Lambda_d}, \qquad C_k = \frac{k+1}{\omega_d}.$$

The PDFs of the received SNRs at the eavesdropper are given by [11] eq. (31)

$$f_{\gamma_e}^{\mathrm{SC}}(\gamma_e) = \sum_{t=0}^{N_e-1} \frac{N_e(-1)^t}{\omega_e\Lambda_e}\binom{N_e-1}{t}\frac{1+\frac{1}{\Lambda_e}+\frac{(t+1)\gamma_e}{\omega_e}}{\left(\frac{1}{\Lambda_e}+\frac{(t+1)\gamma_e}{\omega_e}\right)^2} \times \exp\left(-\frac{(t+1)\gamma_e}{\omega_e}\right) \tag{6}$$

where the constants are defined as

$$A_t = \frac{N_e(-1)^t}{\omega_e\Lambda_e}\binom{N_e-1}{t}, \qquad B_e = \frac{1}{\Lambda_e}, \qquad C_t = \frac{t+1}{\omega_e}.$$

B. Probability of non-zero secrecy capacity (PNZSC)

The PNZSC is expressed as

$$\Pr\left(C_s^{\mathrm{SC}}>0\right) = \Pr(\gamma_d>\gamma_e) = \int_0^\infty f_{\gamma_d}(\gamma_d)\left\{\int_0^{\gamma_d} f_{\gamma_e}(\gamma_e)\,d\gamma_e\right\}d\gamma_d. \tag{7}$$

Using the identities from [12] (3.352.4) and (3.353.3), the closed-form equation of PNZSC can be written as

$$\Pr\left(C_s^{\mathrm{SC}}>0\right) = \sum_{t=0}^{N_e-1}\sum_{k=1}^{N_d} A_t(-1)^{k-1}\binom{N_d}{k}J_{t,k} \tag{8}$$

where

$$J_{t,k} = \frac{1}{\Delta B_e} + \left(\frac{\Delta-D}{\Delta^2}-\frac{G}{C_t\Delta}\right)e^{\frac{GB_e}{C_t}}E_1\left(\frac{GB_e}{C_t}\right) + \frac{D-\Delta}{\Delta^2}e^{\frac{G}{D}}E_1\left(\frac{G}{D}\right) \tag{9}$$

The constants are defined as

$$G = C_t + \frac{k}{\omega_d}, \quad D = \frac{k\Lambda_d}{\omega_d}, \quad \Delta = C_t - B_eD, \quad \Delta\neq 0.$$

C. Intercept Probability (IP)

The intercept probability is given by

$$\Pr\left(C_s^{SC}<0\right) = 1-\int_0^\infty f_{\gamma_d}(\gamma_d)\left\{\int_0^{\gamma_d} f_{\gamma_e}(\gamma_e)\,d\gamma_e\right\}d\gamma_d. \tag{10}$$

Using the identities from [12] (3.352.4) and (3.353.3), the closed-form equation of IP can be written as

$$\Pr\left(C_s^{\mathrm{SC}}<0\right) = 1-\sum_{t=0}^{N_e-1}\sum_{k=1}^{N_d} A_t(-1)^{k-1}\binom{N_d}{k}J_{t,k} \tag{11}$$

D. Secure Outage Probability (SOP)

The SOP is expressed as

$$P_{\mathrm{out}}^{SC}(R_s) = 1-\int_0^\infty f_{\gamma_e}(\gamma_e)\left\{\int_x^\infty f_{\gamma_d}(\gamma_d)\,d\gamma_d\right\}d\gamma_e, \tag{12}$$

where

$$x = e^{2R_s}(1+\gamma_e)-1.$$

Using the identities from [12] (3.352.4) and (4.337.2), the closed-form equation of SOP can be written as

$$P_{\mathrm{out}}^{\mathrm{SC}}(R_s) = 1-\sum_{t=0}^{N_e-1}\sum_{k=1}^{N_d} A_t(-1)^{k-1}\binom{N_d}{k}\frac{e^{-k(\theta-1)/\omega_d}}{M_k}I_{t,k} \tag{13}$$

where

$$M_k = 1+\frac{k\Lambda_d(\theta-1)}{\omega_d}.$$

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The parameter $I_{t,k}$ admits the following closed-form expression:

$$I_{t,k} = \frac{1}{\Delta' B_e} + \left(\frac{\Delta' - D_k''}{(\Delta')^2} - \frac{G'}{C_t \Delta'} \right) e^{\frac{G' B_e}{C_t}} E_1\left(\frac{G' B_e}{C_t} \right) + \frac{D_k'' - \Delta'}{(\Delta')^2} e^{\frac{G'}{D_k''}} E_1\left(\frac{G'}{D_k''} \right) \quad (14)$$

where

$$D_k' = \frac{k \Lambda_d \theta}{\omega_d}, \quad C_k' = \frac{k\theta}{\omega_d}, \quad D_k'' = \frac{D_k'}{M_k},$$
$$G' = C_t + C_k', \quad \Delta' = C_t - B_e D_k''.$$

IV. Numerical Results

The analytical expressions for IP, PNZSC, and SOP derived in Section III are numerically evaluated using MATLAB. These analytical results are then compared with corresponding Monte Carlo simulation outputs to verify the final equations.

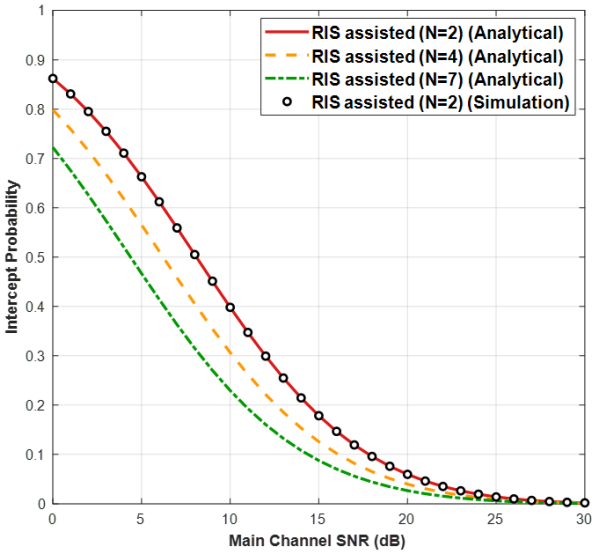


Fig. 2. Effect of N on IP for $N_d = 2$, $N_e = 2$, and $\bar{\gamma}_e = 3\,\text{dB}$.

Figure 2 depicts the variation of intercept probability with respect to the main channel SNR for different RIS element counts, specifically, $N = 2$, 4, and 7. The analysis confirms that intercept probability decreases monotonically with increasing SNR, indicating enhanced secrecy reliability at higher SNR levels. Furthermore, increasing the number of RIS reflecting elements significantly reduces the intercept probability, confirming that a larger RIS array improves the system's physical layer security by limiting the eavesdropper's interception capability. The simulation results for $N = 2$ agree well with the analytical curve, confirming the accuracy of the derived expression in (11).

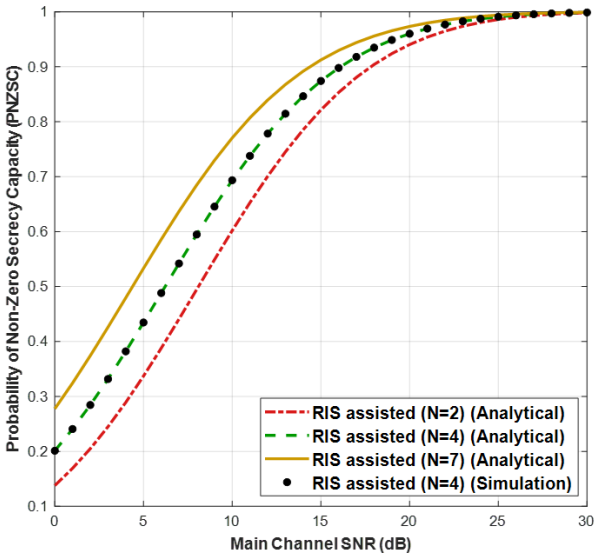


Fig. 3. Impact of N on PNZSC for $N_d = 2$, $N_e = 2$, and $\bar{\gamma}_e = 3\,\text{dB}$.

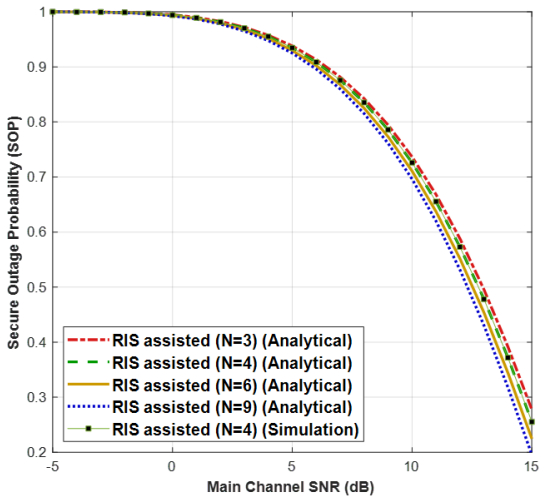


Fig. 4. Impact of N on SOP for $N_d = 2$, $N_e = 2$, and $\bar{\gamma}_e = 3\,\text{dB}$.

Figure 3 illustrates the PNZSC versus the main channel SNR for different RIS configurations with $N = 2$, 4, and 7 reflecting elements. It is observed that the PNZSC increases monotonically with SNR for all configurations, indicating that secure communication becomes more reliable as the main channel quality improves. Moreover, increasing the number of RIS reflecting elements further enhances the PNZSC performance. Among the considered configurations, N = 7 achieves the highest PNZSC, owing to the stronger coherent beamforming gain at the legitimate receiver. The simulation results for N = 4 agree well with the analytical curve, confirming the accuracy of the derived expression in (8).

Figure 4 depicts the SOP versus the main channel SNR for different RIS configurations, specifically N = 3, 4, 6,

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and 9 with secrecy rate threshold $R_s = 0.5$ bits/s/Hz. The results show that the SOP decreases monotonically with increasing SNR, indicating enhanced secrecy performance over the considered range. In addition, increasing the number of RIS reflecting elements further improves secrecy performance, as evidenced by the SOP curve for N = 9, which exhibits the lowest outage probability throughout the entire SNR range. The simulation result for N = 4 agrees well with the analytical curve, confirming the accuracy of the derived expression in (13).

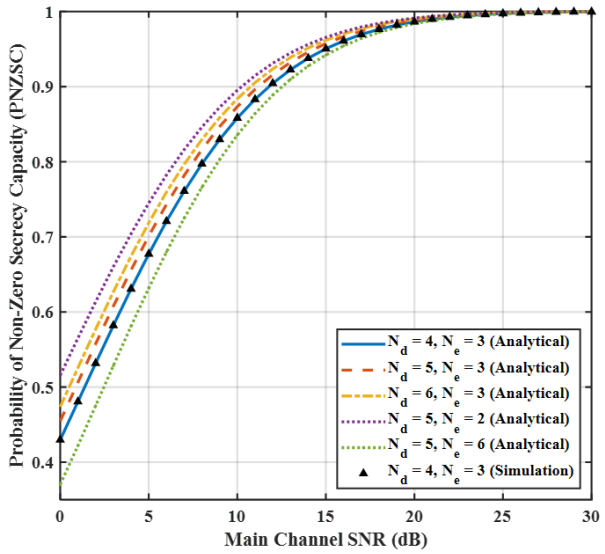


Fig. 5. Variation of PNZSC with respect to N_d and N_e for $N = 4$ and $\bar{\gamma}_e = 3$ dB.

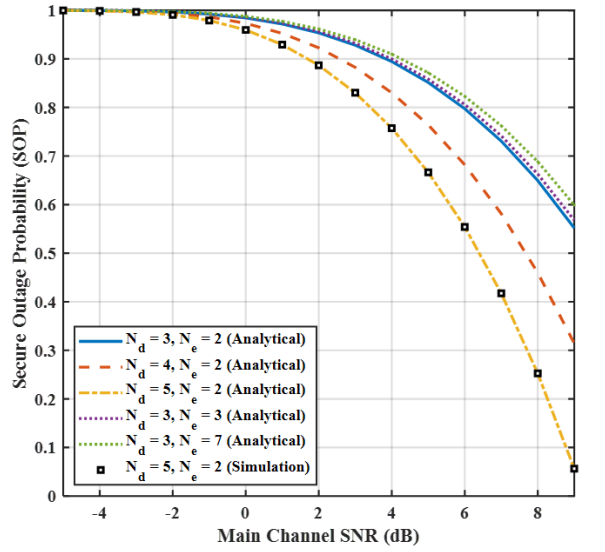


Fig. 6. Variation of SOP with respect to N_d and N_e for $N = 4$ and $\bar{\gamma}_e = 3$ dB.

Figure 5 illustrates the PNZSC versus the main channel SNR for five antenna configurations: $(N_d = 4, N_e = 3)$, $(N_d = 5, N_e = 3)$, $(N_d = 6, N_e = 3)$, $(N_d = 5, N_e = 2)$, and $(N_d = 5, N_e = 6)$. The results show that, for a fixed number of eavesdropper antennas $(N_e = 3)$, increasing the number of legitimate receiver antennas from $N_d = 4$ to $N_d = 6$ consistently improves the PNZSC, confirming the advantage of receiver-side diversity at the legitimate receiver. Conversely, for a fixed number of legitimate receiver antennas $(N_d = 5)$, increasing the number of eavesdropper antennas from $N_e = 2$ to $N_e = 6$ degrades the PNZSC, since additional eavesdropper antennas enhance the interception capability of the wiretap channel. The simulation result for $(N_d = 4, N_e = 3)$ shows close agreement with the analytical curve, validating the accuracy of the derived expression in (8).

Figure 6 depicts the SOP versus the main channel SNR for various antenna configurations (N_d and N_e). The analysis confirms that the SOP decreases monotonically as the SNR increases across all configurations. Furthermore, the data clearly show that, for a fixed number of eavesdropper antennas N_e, increasing the number of legitimate antennas N_d significantly reduces the SOP. Conversely, for a fixed N_d, increasing the number of eavesdropper antennas N_e raises the SOP, thereby degrading the secrecy level. The simulation result for $(N_d = 5, N_e = 2)$ shows close agreement with the analytical curve, validating the accuracy of the derived expression in (13).

V. Conclusion

This work established a unified analytical framework for characterizing outage-based secrecy behavior in RIS-assisted wiretap channels under receiver-side selection combining diversity. The analysis demonstrates that RIS-assisted propagation not only improves the average quality of the legitimate channel but also reshapes the relative SNR distribution between the legitimate receiver and the eavesdropper, directly influencing the achievable secrecy region. The results further reveal an asymmetric impact of receiver diversity on secrecy performance, where diversity gain at the legitimate receiver strengthens secrecy reliability, whereas diversity gain at the eavesdropper increases interception capability. Nevertheless, increasing the number of RIS reflecting elements effectively compensates for stronger wiretap conditions by enhancing the cascaded legitimate channel gain without additional transmit-side complexity. From an implementation perspective, SC diversity achieves a favorable trade-off between secrecy performance and hardware complexity, owing to its single RF-chain requirement. Overall, the presented framework highlights the potential of combining passive RIS beamforming with low-complexity diversity techniques for scalable and energy-efficient secure wireless communication. Future work will investigate multicasting scenarios in RIS-assisted secure networks and extend the analysis to multiple spatially distributed eavesdroppers, where coordinated eavesdropping may require more advanced RIS phase optimization strategies.

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